



UNIVERSITY OF BOLOGNA  
MASTER'S DEGREE IN COMPUTER SCIENCE AND  
ENGINEERING

# Lab-activity-02

*Laboratory Report*

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## 0.1 Objective

The objective of this activity is to program a robot in ARGoS so that it can reach a light source while avoiding collisions with obstacles such as walls, boxes and other robots.

The controller combines two simple behaviours into a single composite behaviour: *phototaxis*, which moves the robot towards the light, and *obstacle avoidance*, which changes the wheel speeds when an obstacle is detected in front of the robot.

The controller is written in Lua and uses the foot-bot light sensors, proximity sensors and differential steering wheels. The wheel velocity is kept below the maximum value of 15, as required by the assignment.

## 0.2 Behavior Description

At the beginning of each step, the robot sets both wheel speeds to the forward value. This means that, by default, the robot moves straight.

Then, the robot reads all the light sensors. The values of sensors from 1 to 12 are summed to represent the light perceived on the left side, while sensors from 13 to 24 are summed to represent the light perceived on the right side. The controller also computes the total amount of light detected by the robot.

If the total light value is greater than the threshold `LIGHT_REACHED`, the robot considers the target reached and stops. This allows the robot to stay close to the light source once it has arrived.

If the robot has not reached the light yet, it checks the proximity sensors. Sensors from 1 to 4 are used for the left-front side, while sensors from 21 to 24 are used for the right-front side. If the sum of these values is greater than `OBSTACLE_CLOSE`, the robot considers that there is an obstacle in front of it.

When an obstacle is detected, the robot turns towards the less occupied side. If the obstacle is stronger on the left side, the robot keeps the left wheel faster and slows down the right wheel, turning away from the obstacle. If the obstacle is stronger on the right side, it does the opposite.

If there is no close obstacle, the robot follows the light. If the light is stronger on the left side, the controller slows down the left wheel and keeps the right wheel faster, so the robot turns left. If the light is stronger on the right side, it slows down the right wheel and turns right. If the difference between the two sides is small, the robot continues moving forward.

The final part of the step checks that the wheel velocities do not exceed `MAX_VELOCITY`, and then sends the selected values to the wheels.

## 0.3 Tests

The controller was tested following the scientific approach suggested in the assignment: first, the expected behaviour was defined, and then the simulation was observed under different conditions.

The first test used a simple arena with the light visible and no difficult obstacle

configuration. In this case, the expected behaviour was that the robot should move towards the brighter side and stop when close to the light. The behaviour matched the expectation.

The second test used an arena with boxes placed between the robot and the light. In this case, the robot was expected to give priority to obstacle avoidance and then continue towards the light after passing the obstacle. The controller was able to avoid frontal obstacles because the proximity behaviour has higher priority than phototaxis.

The controller was also tested with sensor and actuator noise enabled in the ARGoS configuration. With noise, the movement is less regular, but the small tolerance used when comparing left and right light values helps reduce unnecessary oscillations.

Additional tests were considered by changing the number and position of obstacles, and by changing the initial position of the robot. These tests are useful because a controller that works only from one starting position is not very reliable. The same controller can also be tested with different light intensities or with more than one robot in the arena, as suggested in the laboratory text.

## 0.4 Conclusion

The implemented controller is simple and reactive. It does not build a map of the environment and does not use memory: at every step, it directly chooses the wheel velocities from the current sensor readings.

This solution is suitable for the task because the robot has only to combine two immediate behaviours: going towards the light and avoiding close obstacles. The obstacle avoidance behaviour has priority over phototaxis, while stopping near the light has the highest priority. In this way, the robot can search for the target, avoid collisions and remain close to the light source after reaching it.